

INFOGRAPHIC: Non-Parametric Methods

Statistical Modelling Without Distributional Assumptions

[PARAMETRIC]

Assumes a distribution (e.g., Normal). Focuses on parameters like mean (μ) and variance (σ^2).

[NON-PARAMETRIC]

Distribution-free. Focuses on the structure, signs, or **ranks** of the data.

1. Why Use Non-Parametric Methods?

When your data violates classical assumptions, parametric tests can fail or give misleading p-values.

- **No Distribution Assumptions:** Perfect when data is skewed, multimodal, or has heavy tails.
- **Small Sample Sizes:** When you cannot verify if the population is normally distributed.
- **Ordinal Data:** Ideal for data that can be ordered but doesn't have exact numerical distances (e.g., Likert scales, rankings).
- **Robustness:** Not heavily influenced by extreme outliers.

2. The Magic of Ranking: How it Works

Instead of analyzing the raw data values directly, we convert them into their ordinal positions (ranks).

The Transformation Process:

Raw Data: [12.5, 3.1, 89.4, 12.5, 0.2]

Sorted Data: [0.2, 3.1, 12.5, 12.5, 89.4]

Assigned Ranks: [1, 2, 3.5, 3.5, 5] (Ties receive the average rank!)

By throwing away the specific values and keeping only the ranks, we eliminate the impact of scaling and outliers while preserving the relative structure of the dataset.

3. Essential Rank-Based Statistical Tests

For almost every parametric test, there is a non-parametric rank-based equivalent:

Parametric Goal (Assumes Normality)	Non-Parametric Rank Alternative	What it Actually Measures
One-sample t-test / Paired t-test	Wilcoxon Signed-Rank Test	Tests if the median of differences is zero based on sign and magnitude of ranks.

Parametric Goal (Assumes Normality)	Non-Parametric Rank Alternative	What it Actually Measures
Two-sample independent t-test	Mann-Whitney U Test (Wilcoxon Rank-Sum)	Tests whether the distribution of two independent groups is shifted.
One-way ANOVA	Kruskal-Wallis Test	Compares medians across 3 or more independent groups.
Pearson Correlation (r)	Spearman's Rank Correlation (ρ)	Measures monotonic relationships between variables instead of linear ones.

4. Deep Dive: Mann-Whitney U Test

Let's look closely at how independent groups are compared using ranks.

The Formula Structure

For two groups of size n_1 and n_2 , we combine all data, rank them, and sum the ranks for Group 1 (R_1):

$$U_1 = n_1 n_2 + [n_1(n_1 + 1) / 2] - R_1$$

If the null hypothesis is true, the ranks will be mixed randomly between both groups, and U will be close to its expected mean $\mu_U = (n_1 n_2) / 2$.

5. The Trade-Off: Parametric vs. Non-Parametric

Asymptotic Relative Efficiency (ARE):

When assumptions *are* perfectly met, non-parametric tests have slightly less power (typically around 95% efficiency compared to t-tests). However, when assumptions *are violated*, non-parametric tests are vastly more powerful and accurate!